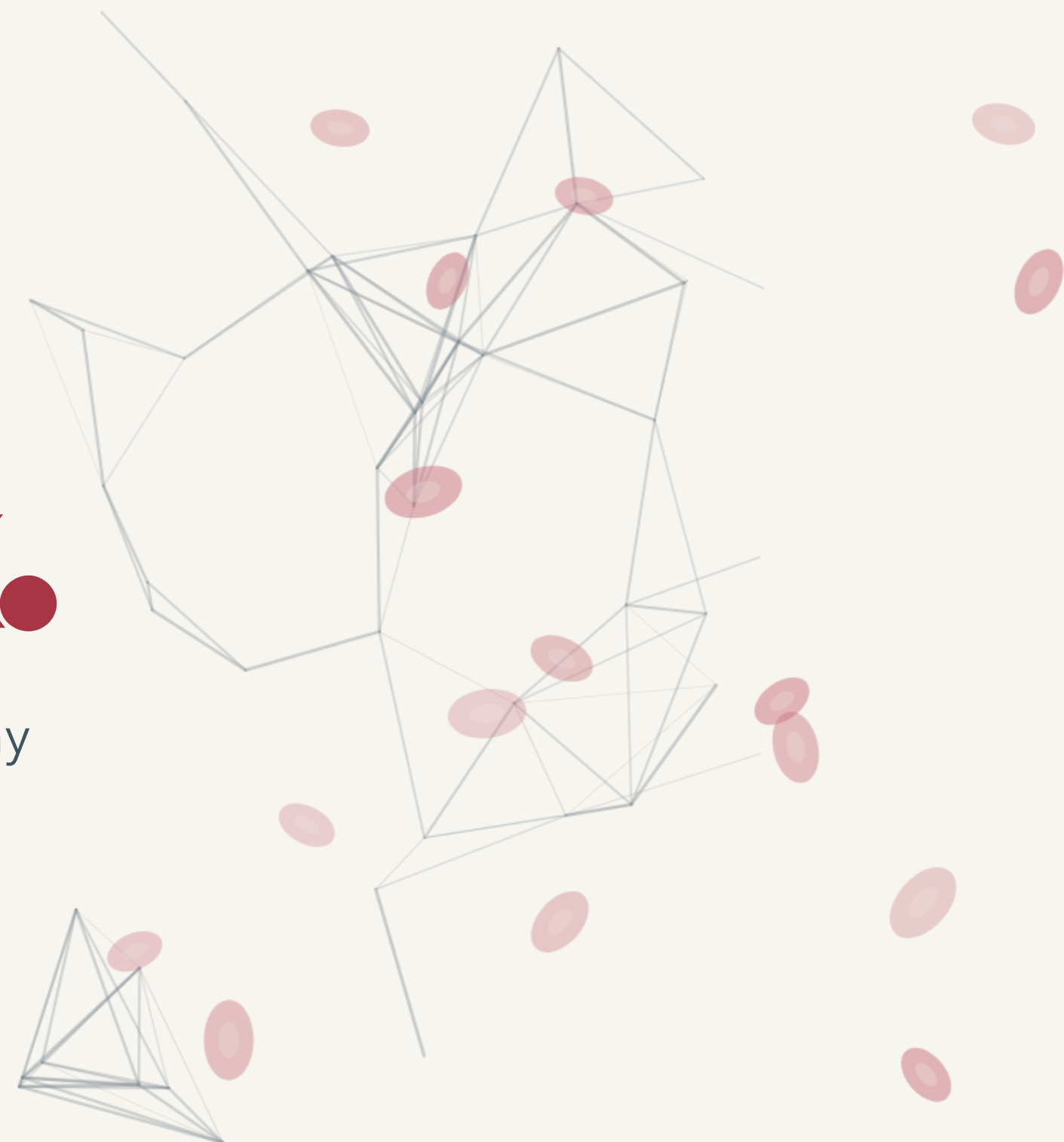


Before you tick the **box.**

Acute clots and thrombophilia testing: when, why
—and will it change what you do?

Dr Abdul Mannan

Consultant Haematologist • Thrombosis Lead, Betsi Cadwaladr University
Health Board



A swollen left leg, and a reflex

An adult with six days of left-leg swelling. No surgery, major immobility or obvious trigger. First event.

Ultrasound: **left common femoral and external iliac vein thrombosis.**

Which result do you need before treating this clot?

None. Today's job is the clot, the limb and the lungs. Not the form.

HAEMATOLOGY REQUEST

URGENT

- ☒ Protein C
- ☒ Protein S
- ☒ Antithrombin
- ☒ Lupus anticoagulant
- ☒ Factor V Leiden
- ☒ Prothrombin G20210A
- ☒ "Full thrombo

One job done

WHAT THE LABORATORY TEXTBOOK SAYS

“Testing during the acute phase of a thrombotic event is **not recommended.**

The clot, acute illness and treatment can all distort the answer.

Dacie and Lewis Practical Haematology · chapter 19 · pre-analytical factors

Five jobs today. The patient comes first.

Life or limb threatened? **Escalate immediately; assess and treat in parallel.**

1

Confirm it

Wells, D-dimer,
ultrasound or CTPA.
Baseline bloods.

2

Name the trigger

Provoked,
unprovoked or
persistent risk? This
informs the later
duration review.

3

Limb and lungs

Is the leg
threatened? Is the PE
high risk?

4

Bleeding risk

What could this
anticoagulant make
bleed?

5

Anticoagulate

Prevent propagation
and embolism. Let
endogenous
fibrinolysis work.

Not on the list:

~~Thrombophilia screen~~

Suspected DVT: the NICE pathway

History and examination, then the **two-level DVT Wells score**

Wells 2 or more • **DVT likely**

Proximal leg vein **ultrasound**, result within **4 hours**

No scan within 4 hours? **D-dimer, then interim therapeutic anticoagulation**, scan within 24 hours

Scan positive
Anticoagulate

Scan negative
Check D-dimer. If positive: stop interim treatment*; repeat scan in 6–8 days.

Baseline bloods: FBC, renal and liver function, PT, APTT. **Do not delay indicated interim anticoagulation** for baseline results. Review within 24 hours.

Wells 1 or less • **DVT unlikely**

D-dimer, result within **4 hours**. Consider an age-adjusted cut-off over 50

No result within 4 hours? Interim therapeutic anticoagulation while waiting

D-dimer positive
Ultrasound within 4 hours, or interim anticoagulation and scan within 24 hours

D-dimer negative
DVT unlikely. Think again and safety-net

Clot confirmed? Start the DVT checklist now. Same questions, every patient, from day one.

NICE NG158 1.1.2–1.1.10, 1.1.14, 1.3.4 • *Unless anticoagulation has another established indication.

One checklist. The same history, every patient.

CHECKLIST FOR CONSULTANT HAEMATOLOGISTV5 • 2023

PATIENT LABEL	DATE	DATE OF DVT MDT
---------------	------	-----------------

CLOT

Site: Left • Right • Iliac • Femoral • Popliteal • Calf • Other
1st VTE • 2nd VTE • Provoked vs Unprovoked

TRIGGER

Recent surgery • Recent immobility • Travel history
OCP or HRT • COVID / recent infection • Family history of VTE

HIDDEN CAUSE

Weight loss • Appetite • Bowel habit • Drenching night sweats
Past cancer: site, how long ago • Diabetes • IBD • COPD • CKD
Smoker / vape • Alcohol units • Weight

BLEEDING

Bleeding disorder or bleeding in last 3 months
Miscarriage, PPH, PV bleed • SSRI / SNRI
Antiplatelet or anticoagulant
Hb • Creatinine clearance • VTE bleed score

PLAN

Anticoagulant • D-dimer • Investigations requested by consultant
Haematology clinic • EOTC 3 or 6 months

- Where is the clot, and is it the first?

Site, side, first or second VTE.
- Provoked or unprovoked?

Surgery, immobility, travel, OCP, infection: the same questions every time.
- Is something hidden?

Cancer symptoms guide targeted tests. No blind screening.
- Safe to anticoagulate, and with what?

Bleeding, antiplatelets, SSRIs, Hb, renal function, bleed score.
- What happens next?

MDT, drug, tests, clinic, review at 3 or 6 months.

Uniform history → clear context → informed treatment review at 3 months.

Provoked or unprovoked? Start with the history.

PROVOKED

A recent transient trigger: major surgery, trauma or immobility. Hormones, pregnancy and puerperium also matter; strength and persistence of the trigger matter.

UNPROVOKED

No recent major clinical risk factor, and not on hormonal therapy. Review history, examination and baseline bloods. No extra cancer tests unless symptoms or signs point to one.

CHECKLIST

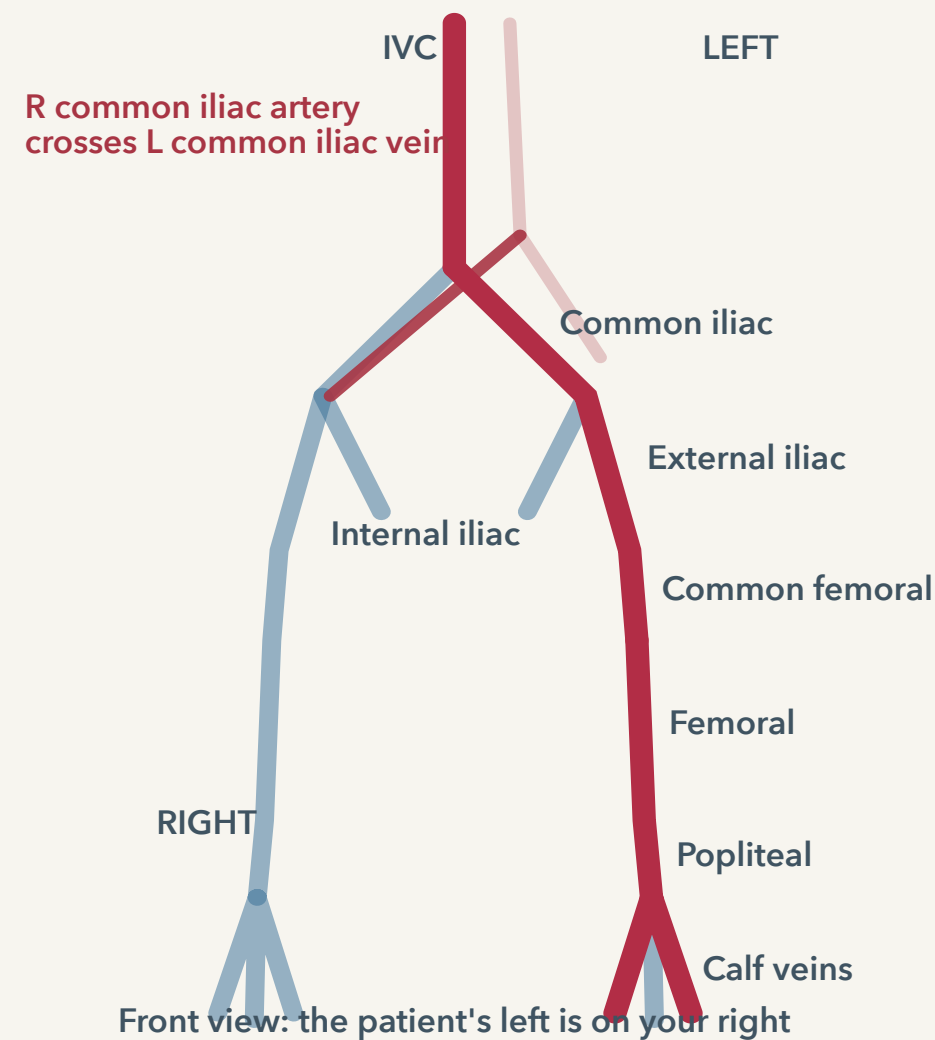
The **Trigger** rows on our DVT checklist capture exactly this, the same way for every patient.

Recurrence risk comes mainly from the **clinical context**, with bleeding risk and patient preference guiding duration—not from an inherited panel.

Teaching summary • BSH 2022 / NICE NG158

Definitions and cancer advice: NICE NG158 glossary, 1.8.1 and 1.8.2

How high does the clot go?



Distal	Calf veins only
Proximal	Popliteal and above. Anticoagulate
Iliofemoral	Common femoral / iliac. Early vascular or IR discussion; consider iliac compression when appropriate.
Iliocaval	IVC involvement. Bilateral swelling or back / abdominal pain can be clues; urgent senior review.

Blue, tense, very painful leg? Phlegmasia can threaten the limb. Start emergency management and call vascular now; do not wait for thrombophilia results.

Selected symptomatic iliofemoral DVT: consider catheter-directed thrombolysis if symptoms <14 days, good function, life expectancy ≥1 year and low bleeding risk.

NICE NG158 1.6.1 • not every iliac clot needs intervention

Phlegmasia (section 2.10) and May-Thurner (pathophysiology): ESVS 2021 venous thrombosis guideline, Kakkos et al.

PE: grade the risk, then match the treatment

High	Cardiac arrest, obstructive shock or persistent hypotension attributable to PE.	Unfractionated heparin infusion. Consider thrombolysis. Call senior and critical care.
Intermediate high	Stable, higher clinical risk, with RV dysfunction + raised troponin .	Anticoagulate and monitor closely. Rescue reperfusion if the patient deteriorates.
Intermediate low	Stable with one or neither marker at higher clinical risk; also low PESI with RV dysfunction / raised biomarkers.	Anticoagulate. Monitor and reassess; disposition follows clinical risk.
Low	Low clinical risk, without adverse RV / biomarker findings; assess outpatient suitability.	Anticoagulate. Consider outpatient treatment with a validated tool.

Routine primary thrombolysis is **not** recommended in intermediate or low risk PE. Thrombolysis is for the patient whose life is at stake.

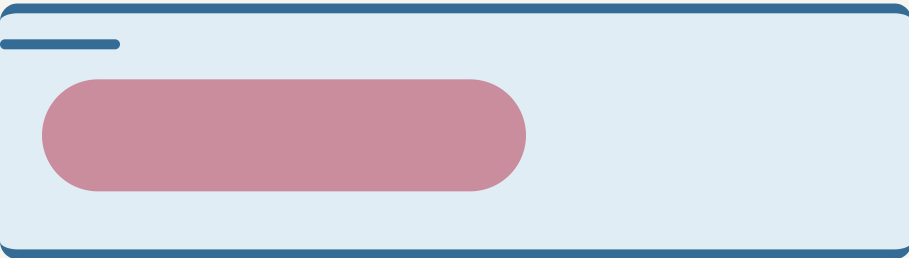
ESC 2019 acute PE guideline, Tables 4 and 8 and recommendations. NICE NG158 1.2.1, 1.3.12, 1.6.2

Stop the growth. Let fibrinolysis work.

Anticoagulants do not directly dissolve the clot.

01 • UNTREATED

The clot grows



Propagation and embolism remain possible.

02 • ANTICOAGULATED

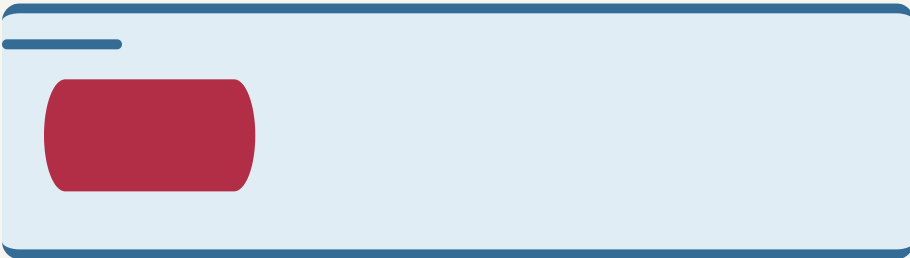
Further clotting suppressed



The growth front is controlled.

03 • ENDOGENOUS FIBRINOLYSIS

The body breaks it down



Recanalisation varies; residual clot can remain.

Treat proximal DVT / PE for **at least 3 months**, then review. Duration depends on recurrence, bleeding and preference.

ESVS 2021 §2.3 • NICE NG158 §1.3–1.4 • Animation is conceptual, not a measured time course.

Know what could bleed. Know what extension looks like.

Before the first dose

- Bleeding in the last 3 months? Ulcer, stroke, recent surgery?
- Hb and platelets. PT and APTT
- Creatinine clearance and liver function: they choose the drug and dose
- Aspirin, clopidogrel, NSAIDs, SSRIs or SNRIs
- Falls, alcohol and body weight inform the individual assessment.

Anticoagulation truly contraindicated? An IVC filter can be considered, with a removal plan agreed before it goes in.

NICE NG158 1.7.2, 1.7.4

Signs the clot is extending

- LEG** Swelling climbing to thigh, groin or buttock. New swelling in the other leg
- IVC** Back, flank or abdominal pain. Both legs swollen
- LIMB** Blue or white, cold, severe pain, numbness, weak pulses
- PE** New breathlessness, pleuritic pain, haemoptysis, tachycardia, syncope

New symptoms on treatment? Confirm recurrence / extension, check adherence, dose and interactions, and involve the treating team. Major bleeding, melaena, severe new headache or collapse also need urgent assessment.

PART TWO

Now, the reflex.

What is in a thrombophilia screen, what an acute clot and its treatment do to the results, and who should be tested at all.

What are we actually looking for?

HERITABLE TRAIT	HOW COMMON	RISK OF A FIRST VTE
Factor V Leiden	About 5% of people of European descent	About 5-fold
Prothrombin G20210A	1% to 2% of Europeans	About 3-fold
Protein C or S deficiency	Under 0.5% each	About 5 to 7-fold
Antithrombin deficiency	Under 0.5%	About 15-fold

BSH thrombophilia testing guideline 2022. Prevalence is much lower outside European populations.

Acquired

- **Antiphospholipid syndrome:** lupus anticoagulant, anticardiolipin, anti-β2GPI
- **PNH:** GPI-deficient clone on flow cytometry
- **MPN:** JAK2, CALR, MPL

The relative risk for a **recurrent** clot is lower than for a first clot, so finding a trait rarely changes management.

BSH 2022

A low—or normal—result can mislead.

TEST	ACUTE CLOT	HEPARIN / LMWH	WARFARIN (VKA)	DOAC
Protein C activity	• Low in DIC or sepsis	↑ False high, clot-based assay	↓ Falls on warfarin	↑ False high, clot-based assay
Protein S free antigen or activity	↓ Falls: recent clot, acute phase	↑ False high, clot-based assay	↓ Falls on warfarin	↑ False high, clot-based assay
Antithrombin activity	↓ Can fall after recent clot	↓ Can fall on heparin	• Not a listed effect	↑ False high: Xa inhibitors in Xa assay, dabigatran in IIa assay

If indicated: assess C/S/AT after ≥3 months of treatment *and* an agreed anticoagulant / assay plan. Never stop treatment simply to send a panel.

BSH 2022 Table 1 · Chromogenic protein C / free protein S antigen avoid some DOAC effects; clinical confounding remains.

The drug is in the tube, too.

LA is detected through a **clotting reaction**—not a simple antibody concentration.



BSH: avoid routine LA testing on anticoagulation or within 3 months of an acute clot. If the answer matters, agree a plan with the laboratory.

aCL / anti- β_2 GPI avoid this direct interference. A positive result still needs context and persistence ≥ 12 weeks apart.

BSH APS 2024 · Favaloro & Pasalic 2023 · Do not omit anticoagulant doses to obtain a test.

Can I trust it? Will it change care?

Defer / plan with the lab

C/S/AT: if indicated, after ≥3 months plus a drug / assay plan. LA: defer during anticoagulation / acute thrombosis.

- Protein C
- Protein S
- Antithrombin
- Lupus anticoagulant

Testable; rarely useful now

DNA is unaffected by anticoagulants. That does not justify routine inherited testing after VTE.

- FV Leiden
- Prothrombin G20210A

Targeted, actionable tests

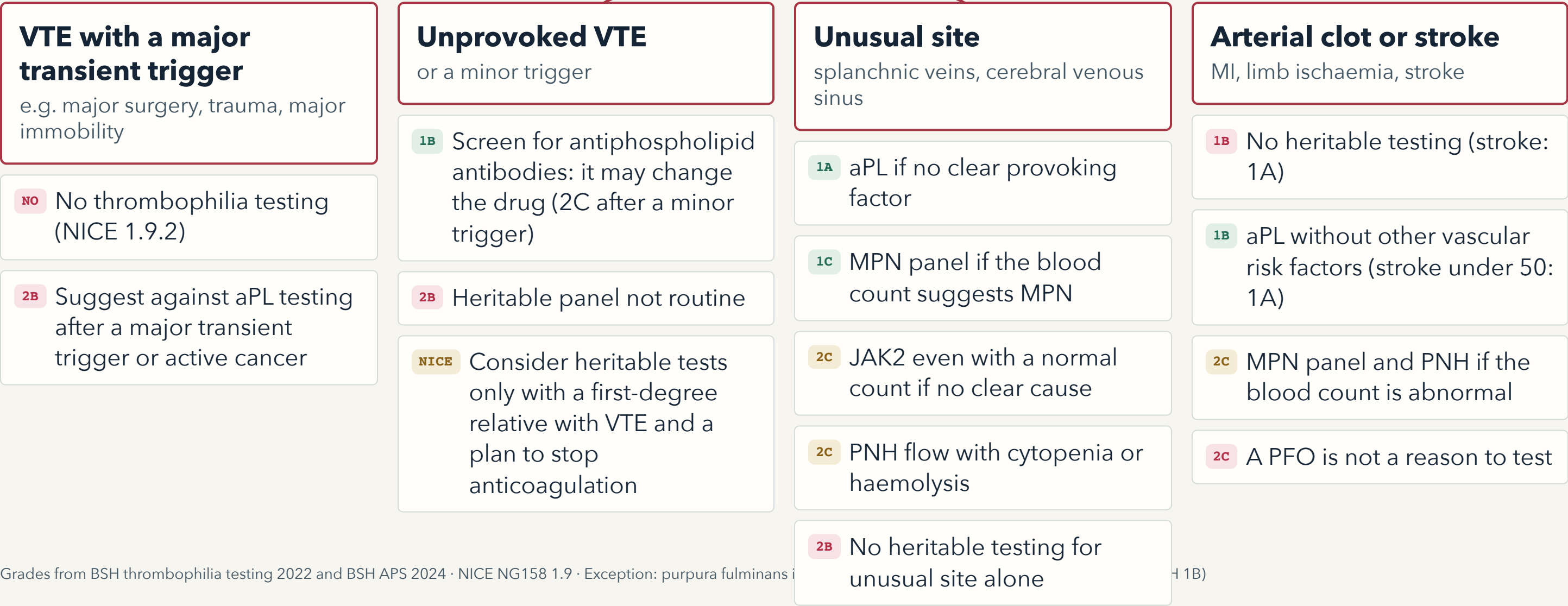
aCL / anti-β₂GPI for an APS question. JAK2 / PNH for the appropriate site and phenotype; confirm and interpret.

- Anticardiolipin
- Anti-β2GPI
- JAK2
- PNH flow

Can be sent is not the same as should be sent.

Start from the clot, not the menu.

Confirmed thrombosis. Which kind?



A wrong result travels further than a right one

FALSE LABEL

"Protein S deficient"

Measured during the clot or on warfarin. Written in the notes, passed to the family.

acute clot / VKA → low PS → lifelong label

FALSE APS

"Lupus anticoagulant positive"

Rivaroxaban or edoxaban in the sample. The patient is moved to warfarin for life, or a real APS is missed.

DOAC → false LA → wrong drug

FALSE COMFORT

"Screen negative, stop at 3 months"

An unprovoked clot with a normal panel still carries the recurrence risk of an unprovoked clot.

normal panel ≠ low risk

WASTE

"Please repeat in 3 months"

Early results are unreliable, so they are repeated. Cost, clinic time and worry for the patient.

test now → repeat later

Routine inherited results usually do not change the **drug, intensity or monitoring** of acute anticoagulation. A specific antithrombin question can be an exception.

Teaching summary · BSH thrombophilia testing guideline, 2022

Six things we still hear on the ward

EVIDENCE

Duration is a clinical decision. A common heterozygous FVL result does not automatically mean lifelong treatment.

BSH 2022

EVIDENCE

Warfarin lowers protein S. Recent clot, pregnancy and oestrogen also confound. Reassess only if useful, with a safe laboratory plan.

BSH 2022 Table 1

EVIDENCE

Not routinely. Selective testing for known familial C, S or AT deficiency may change prophylaxis or reproductive choices.

BSH 2022 • NICE NG158 1.9.5

EVIDENCE

Do not add MTHFR or PAI-1 variants to a routine thrombophilia panel. They do not usefully direct VTE management.

BSH 2022

EVIDENCE

No inherited panel for stroke or PFO alone. Consider aPL in younger patients without conventional vascular risk factors.

BSH 2022

EVIDENCE

An inherited panel is not the answer. Consider an APS question; plan sampling and interpretation with specialist input.

BSH 2022

What we actually did for the patient

TODAY

- Left iliofemoral DVT confirmed. No PE symptoms
- No trigger found: unprovoked. Examined, baseline bloods reviewed
- Leg swollen but pink, warm, pulses present. No phlegmasia
- Low bleeding risk. Therapeutic anticoagulation started
- Under 14 days and fit: vascular / IR asked about catheter-directed treatment and May-Thurner

SENT

- Anticardiolipin IgG and IgM
 - Anti-β2GPI IgG and IgM
- No direct clot-based drug interference. Repeat positive antibodies ≥12 weeks apart; haematology plans valid LA. A single result is not APS.

NOT SENT

Protein C, protein S, antithrombin, lupus anticoagulant, FV Leiden, prothrombin G20210A

AT 3 MONTHS

- Unprovoked: discuss continuing anticoagulation, weighing recurrence against bleeding
- Low bleeding risk: benefits of continuing likely outweigh the risks
- Review once a year if long term

NICE NG158 1.4.3, 1.4.4, 1.4.12 · BSH APS 2024

TAKE HOME

- 1** Life and limb first. Confirm extent, assess context and bleeding, then start appropriate anticoagulation.
- 2** Anticoagulation limits growth and embolism. Endogenous fibrinolysis clears variably; residual clot may remain.
- 3** Acute thrombosis, illness and drugs can distort C/S/AT and LA. Defer routine testing; plan any exception.
- 4** Antibodies, DNA, JAK2 and PNH flow can be tested on anticoagulation. Request only for a defined decision.
- 5** Before any test, ask: **will this change what I do?**

One job done? The job is the patient, not the form.

The clot and the drug rewrite the result

TEST	ACUTE CLOT	HEPARIN / LMWH	WARFARIN (VKA)	DOAC
Protein C activity	• Low in DIC or sepsis	↑ False high, clot-based assay	↓ Falls on warfarin	↑ False high, clot-based assay
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Antithrombin activity	↓ Can fall after recent clot	↓ Can fall on heparin	• Not a listed effect	↑ False high: Xa inhibitors in Xa assay, dabigatran in IIa assay
Lupus anticoagulant DRVVT and APTT	± False positive or negative	± Neutralisation is reagent- and concentration-dependent	± False positive or negative	± False positive or negative; drug / reagent dependent
Anticardiolipin, anti-β2GPI solid phase immunoassay	± Transient antibodies possible	✓ No direct drug effect	✓ No direct drug effect	✓ No direct drug effect
FV Leiden, prothrombin G20210A DNA (PCR)	✓ No direct drug effect	✓ No direct drug effect	✓ No direct drug effect	✓ No direct drug effect
JAK2 V617F • PNH clone PCR on blood • flow cytometry	✓ No direct drug effect	✓ No direct drug effect	✓ No direct drug effect	✓ No direct drug effect

↓ real or misleading fall ↑ false rise, can hide a deficiency ± false positive or negative ✓ not affected

BSH 2022 Table 1 • BSH APS guideline 2024 • Favaloro and Pasalic 2023

Sources behind the decisions

- 1 BSH thrombophilia testing · Arachchillage et al., 2022
- 2 BSH antiphospholipid syndrome · Arachchillage et al., 2024
- 3 NICE NG158 · VTE diagnosis, treatment and testing
- 4 Dacie and Lewis · Investigation of a Thrombotic Tendency, chapter 19
- 5 Favaloro & Pasalic · Laboratory testing for aPL, 2023
- 6 ESVS venous thrombosis guideline · Kakkos et al., 2021
- 7 ESC acute PE guideline · Konstantinides et al.
- 8 Tavasoli et al. · Laboratory diagnosis of protein S deficiency, 2025